"Equity" in Public Health Discourse

Farnoosh Memari and Sebine Scaria
Master in Interdisciplinary Data Science, Duke University
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Dr. Astrid Giugni
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Literature Review

The concept of *equity* in public health discourse is not semantically stable. Rather than having a single, fixed definition, it operates across multiple conceptual registers. Braveman and Gruskin (2003) show that equity functions both as a normative principle, grounded in ideas of justice, and as an empirical construct that must be operationalized through measurable disparities. They define health equity as the absence of systematic differences between socially advantaged and disadvantaged groups, distinguishing it from equality. Braveman (2003) further demonstrates that even technical decisions about how to measure equity are not neutral. For example, the World Health Organization's shift toward measuring inequalities between ungrouped individuals rather than socially defined groups had the effect of obscuring structural inequalities related to race and class. Together, this work shows that equity is both ethically charged and technically constructed, making it difficult to interpret consistently across texts.

Recent advances in natural language processing provide tools for analyzing this kind of semantic variability.

Devlin et al. (2019) introduce BERT, a bidirectional Transformer model that represents words based on their full sentential context rather than assigning them a single fixed meaning. This approach allows the same word to take on different representations depending on how it is used, which is crucial for analyzing concepts like equity. Wiedemann et al. (2019) show that these contextualized embeddings capture polysemy, with different senses of a word forming distinct clusters in embedding space. Their findings demonstrate that semantic distinctions can be recovered computationally, even for concepts with multiple meanings.

At the same time, meaning in complex texts is not only a matter of linguistic context but also of institutional context. Named entity recognition (NER) provides a way to connect language to the actors and structures that shape it. As Ehrmann et al. explain, NER is particularly useful in historical and policy corpora because it transforms unstructured text into structured data by identifying entities such as organizations, locations, and individuals. This allows researchers to move beyond isolated word meanings and examine how concepts are embedded within networks of institutions and actors. In this sense, NER complements embeddings-based approaches by grounding semantic variation in real-world contexts.

Taken together, these approaches suggest that meaning in complex texts is both context-dependent and structured. Contextual embedding models such as BERT capture how meanings shift across linguistic environments, while NER reveals how those meanings are tied to specific institutions and actors. For a concept like equity, this combination is especially important, as its meaning varies not only across sentences but also across domains of use. Computational methods therefore make it possible to treat semantic variation not as noise, but as a key feature of how meaning operates in public health and policy discourse.

Concept and Data Preparation

The conceptual keyword for this project is "equity." All occurrences of the term were extracted across the corpus at the sentence level, which is the primary unit of analysis. To preserve local context, each sentence is also paired with a ±1 sentence window, meaning the sentence before and the sentence after are retained for interpretation. The final corpus contains 27,365 sentences, of which 4,061 contain "equity." These occurrences are distributed unevenly across document categories: 2,203 in state/local documents, 1,036 in federal policy documents, 630 in NGO/nonprofit documents, 140 in academic documents, and 52 in news/commentary documents. Two initial patterns are immediately visible. First, equity language is concentrated in policy and institutional documents, especially state/local and federal sources. Second, although equity is a central concept for the project, it appears in only a minority of sentences overall, which suggests that it is meaningful but not dominant across the corpus.

Named Entity Recognition

While embedding models capture semantic variation, Named Entity Recognition (NER) allows us to identify the institutional actors through which equity is articulated. To examine how the corpus discusses the concept of equity, spaCy NER was applied to all equity-related sentences in the corpus. The primary focus was placed on organizations (ORG) and geopolitical entities (GPE), since these entity types are the most informative for identifying the institutions and locations through which equity is framed in public health discourse. PERSON entities were also extracted, but they were interpreted more cautiously because person labels are less reliable in technical and policy-oriented text.

The corpus was first filtered to sentences containing equity, equitable, inequity, or inequitable, resulting in 4,406 equity-related contexts. From these contexts, 9,934 entity mentions were extracted, corresponding to 5,315 unique normalized entities. In total, 3,191 equity-related sentences contained at least one named entity.

Table 1.
Frequency and percentage distribution of named entity labels in equity-related contexts.

entity_label	Mention	share_pct
ORG	6164	48.24
GPE	888	6.95
PERSON	1060	8.29

The overall frequency distribution shows that ORG is by far the most common entity label, with 6,161 mentions (62.02%), followed by PERSON with 1,059 mentions (10.66%) and GPE with 887 mentions (8.93%). This pattern indicates that equity is discussed primarily through institutions, agencies, and organizational structures, rather than mainly through individuals or geographic references.

When the results are broken down by document type, policy documents account for the highest raw counts, with 4,992 ORG mentions, 706 GPE mentions, and 687 PERSON mentions. NGO documents contain 1,078 ORG mentions, 167 GPE mentions, and 336 PERSON mentions. Academic and news documents contribute much smaller totals overall. Because policy documents make up the largest portion of the corpus, raw frequencies alone are not sufficient for interpretation, and normalized comparisons are therefore more appropriate.

Figure 1. *Overall distribution of named entity labels in equity-related contexts by mention frequency.*

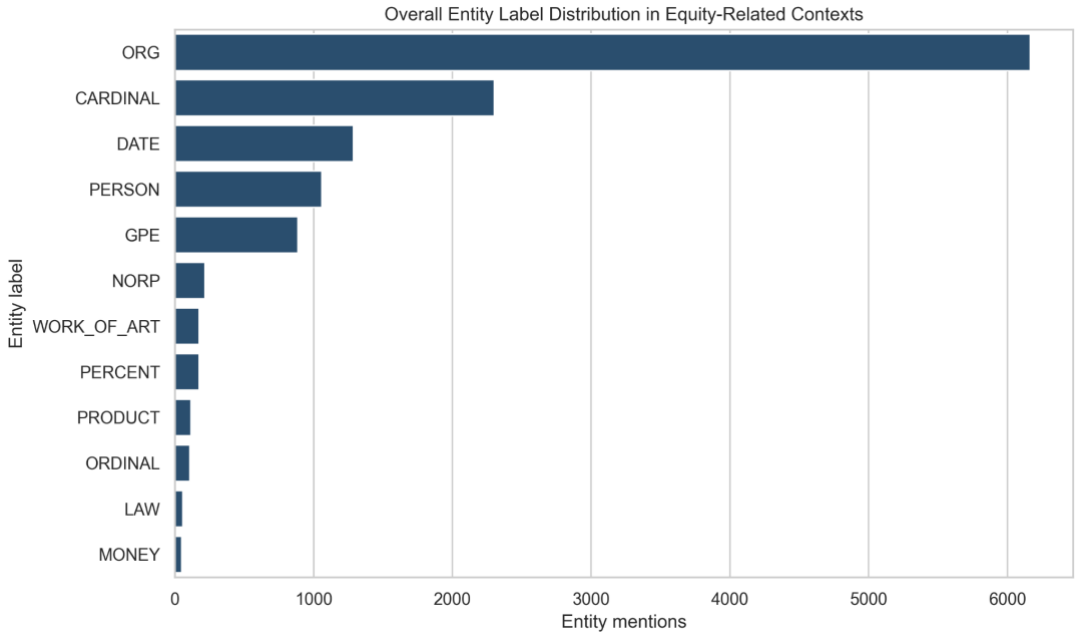
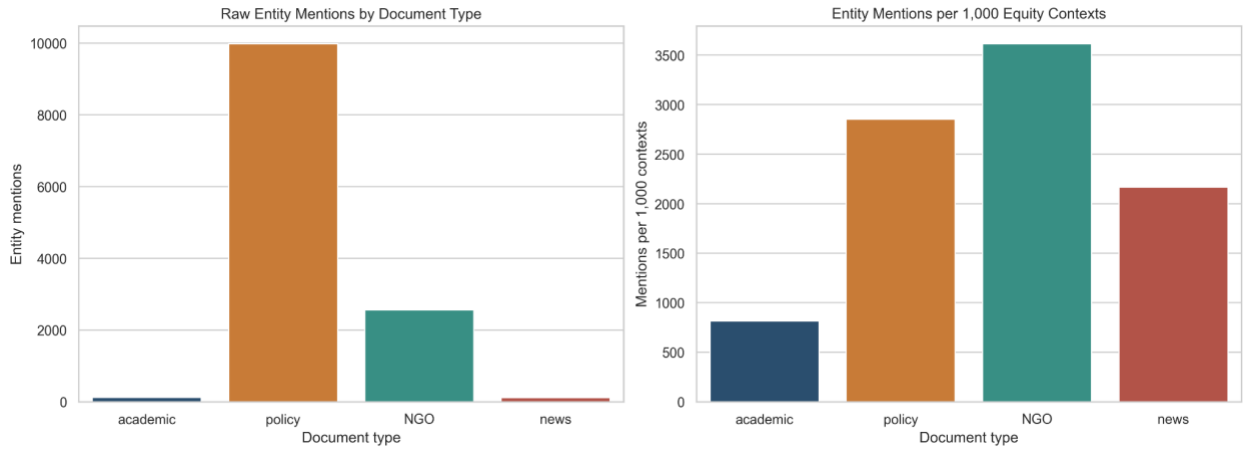


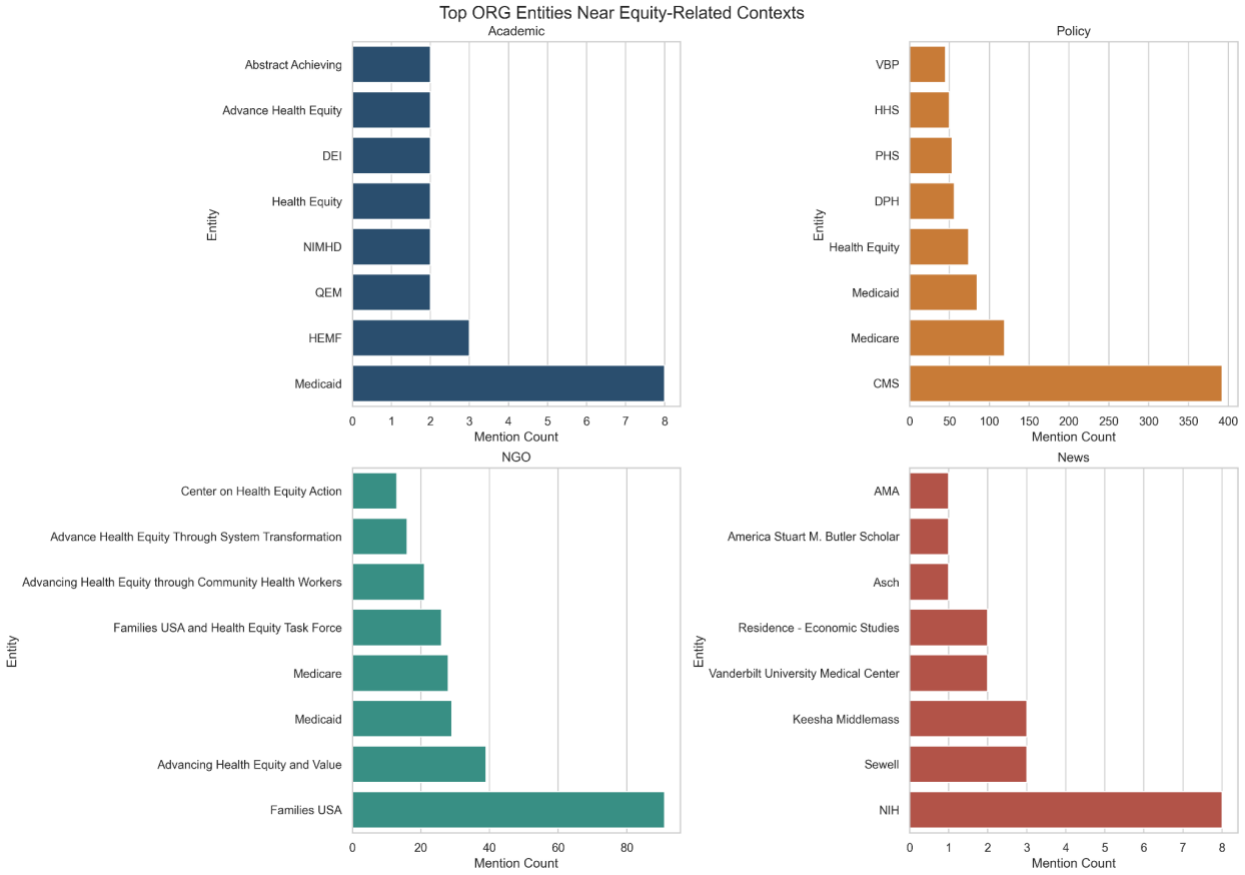
Figure 2. *Raw and normalized entity mentions by document type, showing total counts and mentions per 1,000 equity contexts.*



The comparison between raw and normalized rates shows that policy texts produce the highest raw number of entity mentions, while NGO texts produce the highest mentions per 1,000 equity contexts. This suggests that policy discourse dominates in volume, but NGO discourse is more densely packed with named institutional and geographic references in each equity-related context.

Among the focus entities, the ORG results are especially informative. In policy texts, the most frequent organizations include CMS (392 mentions), Medicare (119), Medicaid (85), HHS (50), and CDC (38). These results suggest that equity is framed in policy documents through public agencies, government programs, and formal administrative systems. In NGO texts, the most frequent organizations include Families USA (91 mentions), Medicaid (29), Medicare (28), and several advocacy-oriented organizational names related to health equity. In this document type, equity appears to be framed more strongly through advocacy networks, coalition work, and reform-oriented institutions.

Figure 3. *Top co-occurring organization (ORG) entities across document types in equity-related contexts by mention count.*



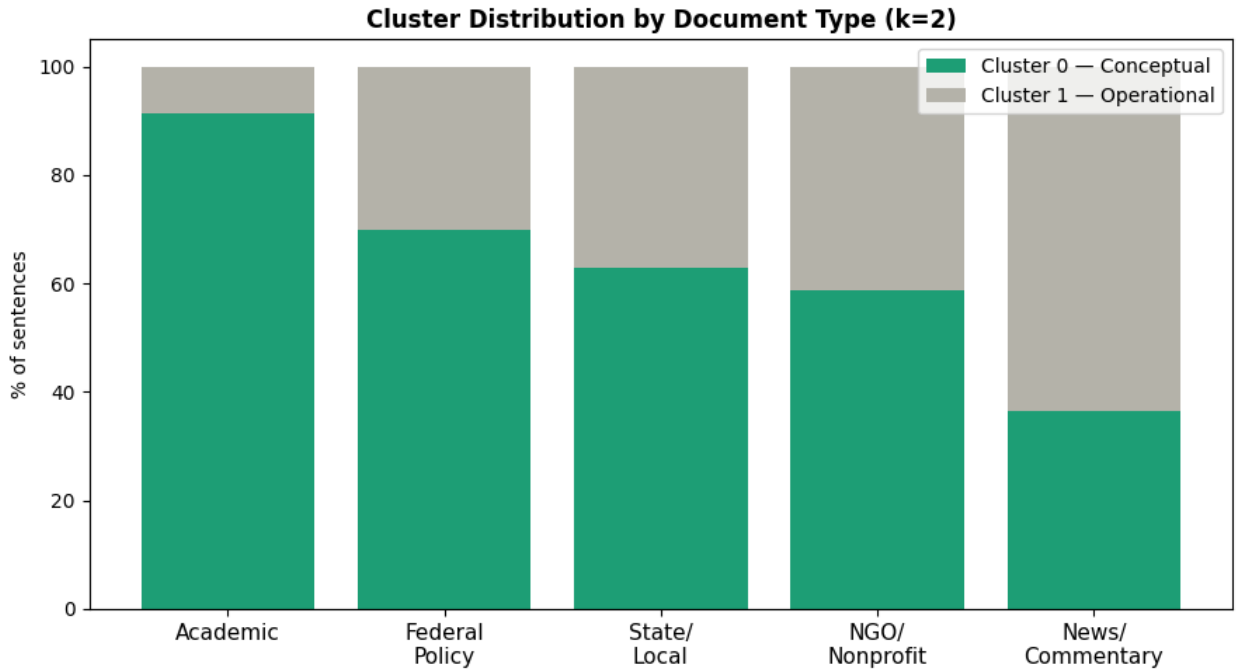
Examples of co-occurrence further support these patterns. In policy documents, equity frequently appears alongside organizations such as CMS, Medicare, Medicaid, HHS, and CDC, indicating that the term is often embedded in the language of governance, administration, and health service delivery. In NGO documents, equity co-occurs with organizations such as Families USA and other advocacy-related groups, suggesting a stronger emphasis on reform, partnership, and community-oriented action. Geographic co-occurrences such as King County, Michigan, and North Carolina further show that equity is often discussed through place-based implementation and jurisdiction-specific policy efforts.

Overall, the NER results suggest that equity is discussed less as an abstract idea and more through the institutions and places that structure public health action. ORG emerges as the strongest focus entity because it captures the administrative and organizational infrastructure of equity discourse across the corpus. GPE is also highly informative because it identifies where equity is being localized and implemented. By contrast, PERSON entities are less central to the interpretation and should be treated cautiously, as the analysis indicates that sentence-level NER in policy text can misclassify programs or acronyms as people.

Contextual Meaning with BERT

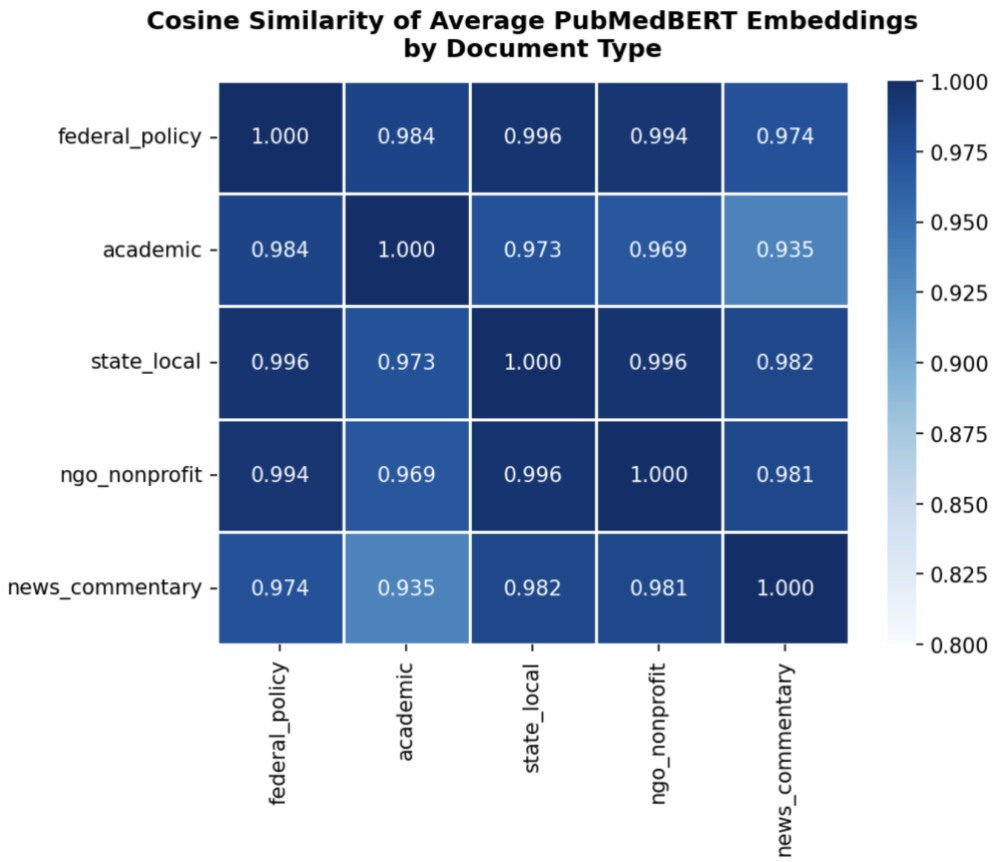
After establishing the institutional context of equity through NER, we next examine how its meaning varies semantically using contextual embeddings. We generated contextual word embeddings using PubMedBERT, a BERT model pre-trained on biomedical text. We chose it because all five document types in our corpus are based on health-specific terminology and this BERT model happened to produce better cluster structures. We used the same sentence design window for each of the 4060 sentences by adding the preceding and following sentences. This design choice provides the model with sufficient context to produce semantically informed representations of 'equity' in each occurrence. We tokenized each context window using WordPiece tokenization, indexed the token positions of the word, passed the full sequence through all 12 BERT layers, and extracted the embeddings at the equity token position by averaging the outputs of the final four hidden layers following Devlin et al. (2019), which identifies this a standard strategy for producing stable contextual representations. Each embedding is a 768-dimensional vector. Of the 4060 contexts, 4040 produced valid embeddings and the rest 20 were excluded since those instances didn't resolve 'equity' as a distinct token, a known edge case of subword tokenization. We applied K-Means Clustering to the L2-normalized embeddings to identify semantic clusterings among equity usages. The optimal number of clusters was selected by calculating the silhouette score which produced k = 2 as the optimal number of clusters with a score of 0.184. This is a better score compared to the silhouette scores of other k values. This is also because the usage of 'equity' in this corpus does not divide into two cleanly discrete categories, but varies along a continuum, with two clusters capturing central tendencies rather than hard boundaries due to which the low-confidence classification predictions is 14.8% .

Figure 4.
K-means cluster distribution across five document types (k=2, silhouette score=0.184). Cluster 0, characterized by coherent, conceptually developed equity language, dominates academic texts (91.4%) and constitutes the majority cluster across federal policy (70.0%), state/local (62.8%), and NGO/nonprofit (58.7%) documents. News/commentary is the only category where Cluster 1 — operationally fragmented equity language — is the majority (63.5%), consistent with its journalistic register and higher structural framing rate in the classification analysis.



Cluster 0 is characterized by coherent, conceptually developed equity language. Academic sentences dominate this cluster by 91.4%. Cluster 1 has more operationally fragmented equity language, sentences where equity co-occurs with measurement vocabulary and accountability contexts. The asymmetry in definition sentence distribution shows that when authors in this corpus explicitly stop to define equity, they do so predominantly in Cluster 0.

Figure 5.
Pairwise cosine similarity between average PubMedBERT equity embeddings across five document types. All values range from 0.935 to 0.996, confirming that equity operates within a shared semantic domain across institutional contexts. Federal policy and state/local documents are most similar (0.996), consistent with their near-identical distributional framing rates. Academic and news/commentary documents are least similar (0.935), reflecting their distinct registers and discursive conventions.



The Pairwise Cosine Similarity matrix tells us that the document types range from 0.935 to 0.996 confirming that all five documents use equity within a shared semantic domain. Federal policy and State/Local documents are semantically closest (0.996), due to their nearly identical distributional framing rates in the classification analysis. The lowest similarity is between Academic and NEWS Documents (0.935). This tells us that the highest cosine similarities establish a shared semantic core - all document types are talking about the same concept, but the analyses from the clusters present variation within that core in the sentence-level.

Supervised Classification (Logistic Regression)

To formalize these semantic distinctions, we trained a logistic regression classifier to distinguish between structural and distributional framings. Class A (Structural/Conceptual framing) captures equity as a social justice related word like justice, rights, structural, systemic, root causes, inequity, and so on. Class B (Distributional/Operational framing) captures equity as an administrative target achieved through programs like data, plan, cms, programs and so on. Labels were assigned via keyword-based weak labeling. Of the 4060 equity sentences, 2762 (68%) received clear labels out of which 337 were structural and 2425 were distributional with 1298 flagged as ambiguous and excluded from training. This imbalance is meaningful since it reflects the corpus's institutional composition where government documents dominate and mainly talk about equity in operational terms.

Table 2.
Comparison of feature configurations for the logistic regression classifier evaluated using 5-fold cross-validation. TF-IDF alone achieves the highest mean F1 score (0.954), outperforming both the combined TF-IDF + BERT and BERT-only configurations, indicating that structural vs. distributional equity framing is primarily a lexical distinction.

Feature Configuration	Mean F1 (5-fold)	Std
TF-IDF	0.954	0.006
TF-IDF + BERT	0.890	0.004
BERT only	0.866	0.009

The top TF-IDF coefficients confirm that the classifier has learned a good amount of meaningful distinctions rather than spurious correlations. The three strongest are data (3.88), plan (3.11), cms (2.79). The appearance of cms as the third-strongest corroborates with the NER finding from Section 2: sentences mentioning the Centers for Medicare and Medicaid Services are overwhelmingly distributional, 379 of 388 CMS sentences being predicted as Class B. This convergence across the three methods suggests that organizational actors are not just co-occurring with equity, but with systematically different framings of it.

Table 3.
Predicted equity framing by document type, applied to all 4,060 equity sentences using the TF-IDF logistic regression classifier. Structural framing rates follow a clear gradient from news/commentary (~32.7%) to federal policy (10.3%), with academic, NGO, and state/local documents falling in between. Values marked with ~ are approximations based on raw count proportions. Rows are sorted by structural framing rate in descending order.

Document Type	Equity Sentences	Structural	Distributional	% Structural
NEWS/ Commentary	52	17	35	32.7%
Academic	140	34	106	24.3%
NGO/Non-profit	630	67	563	10.6%
State/Local	2,202	361	1841	16.4%
Federal Policy	1036	107	929	10.3%

This four-way comparison shows that News and Academic texts carry the highest structural rates. NGO documents take a middle position, consistent with their dual role as advocacy organizations operating within and against institutional policy frameworks. Federal and

State/Local government documents are overwhelmingly distributional, reflecting the compliance-oriented contexts in which their equity language is produced.
When it comes to model performance and results, it is also important to consider model limitations. 599 sentences (14.8%) received low-confidence predictions (below 0.65), which are concentrated in state/local (344) and federal policy (105). These are not just classification errors, but represent the linguistic characteristic of the corpus. An example of a sentence which falls into the low-confidence prediction category is 'The full definition proposed by RWJF is "Health equity means that everyone has a fair and just opportunity to be as healthy as possible.' This sentence has structural signals (fair and just opportunity), but is deployed in a policy compliance context which falls into a distributional framing. The classifier's uncertainty comes from borderline sentences and that happens with sentences that come from government documents because these documents blend together structural equity with distributional framing since these are administrative documents.

Synthesis Section

Across all three methods, the analysis shows that the word *equity* does not have a single stable meaning in public health discourse. Although all five document types appear to reference the same general concept, something confirmed by the high similarity of PubMedBERT embeddings, two distinct underlying framings emerge. The structural framing treats equity as a social justice goal requiring systemic transformation, while the distributional framing treats it as an administrative target achievable through program delivery and data accountability. These two meanings are not randomly distributed, but are organized along institutional lines, with different actors and document types consistently aligning with one framing over the other.

Two primary meanings of equity therefore exist in this corpus. The structural meaning is centered on justice, root causes, and systemic change. The distributional meaning is centered on measurement, programs, and administrative accountability. These meanings are structured by institutional context, shaping what equity demands from organizations and what counts as progress toward it.

The NER results show that equity is not discussed in the abstract, but through institutions. In policy documents, equity most frequently co-occurs with government agencies such as CMS, HHS, and Medicaid, whose roles are administrative and programmatic. In NGO documents, organizations such as *Families USA* are more prominent, reflecting a focus on advocacy and systemic reform. Who invokes equity reliably signals how it is being framed, linking semantic meaning to institutional actors.

The BERT clustering analysis shows that this institutional pattern also has a semantic dimension. Academic sentences are concentrated in the cluster characterized by more coherent and developed equity language. This same cluster contains the majority of sentences where equity is

explicitly defined. By contrast, the operational, measurement-focused cluster draws more heavily from policy and news documents. Equity is therefore not only framed differently across document types, but also embedded in distinct semantic environments.

The classifier reproduces this pattern independently. Without access to document type, it identifies structural framing most often in news and academic texts, and distributional framing in federal and state/local policy documents, with NGO texts falling in between. This reinforces the conclusion that the distinction between the two meanings is systematic and generalizable.

One of the most important findings is that TF-IDF outperforms BERT for the classification task. This suggests that the distinction between structural and distributional framings is primarily encoded in vocabulary rather than subtle contextual variation. Institutions in this corpus do not gradually shift the meaning of equity depending on context; instead, they deploy distinct word sets associated with different institutional registers. The framing of equity is therefore not only semantic, but also discursive.

Approximately 14.8% of equity-related sentences receive low-confidence predictions, reflecting cases where both framings are present within the same sentence. This occurs most often in policy documents, where terms such as “fair,” “just,” and “rights-based” appear alongside administrative and programmatic language. These borderline cases show that the boundary between the two framings is not fixed, but can be blended within institutional discourse.

Limitations

Four limitations should be noted. First, weak labeling may introduce noise because keyword heuristics can misclassify some sentences. Second, the NER analysis used *en_core_web_sm*, a general-purpose model that may miss domain-specific entities, and no entity linking was performed. Third, the corpus provides only a synchronic snapshot and cannot capture changes in equity framing over time or in response to policy events. Fourth, because the distribution of different equity meanings across sources was unknown, the balancing strategy was approximate and may not reflect their true prevalence in the corpus. As a result, some meanings may be overrepresented or underrepresented in the analytical sample. Future work could use manual annotation to validate labels and improve reliability.

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